# IF440: Artificial Intelligence

## Topic 04: Bayesian Theorem

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# Course Sub Learning Outcomes (Sub-CLO):

SubCLO-0221: Students are able to apply knowledge-based reasoning, probabilistic reasoning, and soft computing (C3).

# Review

1. First-Order Logic
2. Syntax of FOL
3. Quantifiers
4. Assertions and Queries
5. Inference in FOL

# Outline

## Pre-Class Assignment

- Veritasium `https://youtu.be/R13BD8qKeTg?si=ghWA3s_duvEbTo6g`

# Using uncertain knowledge

# USING UNCERTAIN KNOWLEDGE

- Suppose you are trying to determine if a patient has appendicitis. You observe the following symptoms:
  1. The patient has **fever**
  2. The patient experienced **vomiting**
  3. **Lower right abdomen will be sore** when pressed, and the pain will be felt immediately after the pressure is released.

# USING UNCERTAIN KNOWLEDGE

We are not 100% certain that the patient has appendicitis because of those symptoms. We are dealing with uncertainty!

# USING UNCERTAIN KNOWLEDGE

- Now suppose you order a CT scan and observe that the patient has a swelling appendix.
- Your belief that the patient is suffering from appendicitis is now much higher.

# USING UNCERTAIN KNOWLEDGE

- In the prev. slides, what you observed affected your belief that the patient is suffering from appendicitis.

- This is called reasoning with uncertainty.

- Wouldn't it be nice if we had some methodology for reasoning with uncertainty?

# Probability

# WHY PROBABILITY ?

- There is lots of uncertainty about the world, but agents still need to act.
- Acting is gambling: agents who don't use probabilities will lose to those who do.
- Probabilities can be learned from data.
- Bayes' rule specifies how to combine data and prior knowledge.

# PROBABILITY

- Probability is an agent's measure of belief in some proposition — **subjective probability**.
- An agent's belief depends on its prior assumptions and what the agent observes.

# NUMERICAL MEASURES OF BELIEF

- Belief in proposition, *f* , can be measured in terms of a number between 0 and 1 — this is the probability of *f* .
  - The probability *f* is 0 means that f is believed to be definitely false.
  - The probability *f* is 1 means that f is believed to be definitely true.
- *f* has a probability between 0 and 1.
- Probability is a measure of an agent's degree of uncertainty.

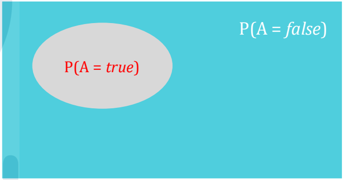
# RANDOM VARIABLES

- Random variables is the basic element of probability.
- Variables in probability theory are called random variables (names begin with an uppercase letter).
- Has a domain – the set of possible values it can take on.
  - E.g. *Weather* to be {*sunny*, *rain*, *cloudy*}

- Boolean random variable has domain {true, false} □ values are always lowercase.
- By convention, $A = true$ is abbreviated simply as $a$, while $A = false$ is abbreviated as $\neg a$.

# RANDOM VARIABLES

- $P(A = true)$ □ the probability that $A = true$.
- $P(A)$ □ the probability of event $A$.
- Also called unconditional/marginal probability.
- Ex.: probability that a card drawn is red $P(red) = 0.5$




The sum of the red and blue areas = 1

# JOINT PROBABILITY

- $P(A \wedge B) \rightarrow$ "Probability of event A and B occurring." Probability of the intersection of two or more events.
- Ex.: the probability that a card is four and red $P(four \wedge red) = \dfrac{2}{52}$

# Conditional Probability

# CONDITIONAL PROBABILITY

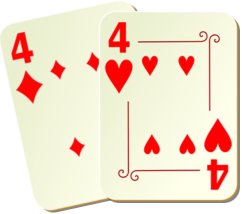
- $P(A|B)$ → "Probability of A conditioned on B" or "Probability of A given B".
- Ex.: given that you drew a red card, what's the probability that it's four

$$P(four|red) = \frac{2}{26}.$$

- In general,

$$P(A|B) = \frac{P(A \wedge B)}{P(B)}$$

$$P(A \wedge B) = P(A|B)P(B)$$



# CONDITIONING

- Probabilistic conditioning specifies how to revise beliefs based on new information.
- An agent builds a probabilistic model taking all background information into account. This gives the prior probability.
- All other information must be conditioned on.
- If evidence $e$ is all the information obtained subsequently, the conditional probability $P(h|e)$ of $h$ given $e$ is the posterior probability of $h$.

# PROBABILITY DISTRIBUTION

- Probabilities of all the possible values of a random variable.
- They should sum to 1.
  - $P(sunny) = 0.7$
  - $P(rainy) = 0.1$
  - $P(cloudy) = 0.2$
- But, as abbreviation: $\mathbf{P}(Weather) = \langle 0.7, 0.1, 0.2 \rangle$
  - **P** statement defines a probability distribution for the random variable *Weather*
  - **P** notation is also used for conditional distributions: $\mathbf{P}(X | Y)$ gives the values of $P(X=x_i | Y=y_j)$ for each possible *i,j* pair.

# Joint Probability Distribution

# JOINT PROBABILITY DISTRIBUTION

- Distributions on multiple variables.
- Probabilities of all combinations values.
- They should sum to 1.

| Weather | Temperature | Probability |
|---|---|---|
| sunny | hot | 150/365 |
| sunny | cold | 50/365 |
| cloudy | hot | 40/365 |
| cloudy | cold | 60/365 |
| rainy | hot | 5/365 |
| rainy | cold | 60/365 |

# JOINT PROBABILITY DISTRIBUTION

- Once you have the joint distribution, you can do anything
- $P(\text{sunny} \vee \text{hot}) = (150 + 50 + 40 + 5)/365$

| Weather | Temperature | Probability |
|---|---|---|
| sunny | hot | 150/365 |
| sunny | cold | 50/365 |
| cloudy | hot | 40/365 |
| cloudy | cold | 60/365 |
| rainy | hot | 5/365 |
| rainy | cold | 60/365 |

# JOINT PROBABILITY DISTRIBUTION

- The product rules for all possible values of *Weather* and *Temperature* can be written as single equation:

$$\mathbf{P}(Weather \wedge Temperature) = \mathbf{P}(Weather|Temperature)\mathbf{P}(Temperature)$$

$$P(sunny \wedge hot) = P(sunny|hot)\ P(hot)$$

$$P(sunny \wedge cold) = P(sunny|cold)\ P(cold)$$

$$P(cloudy \wedge hot) = P(cloudy|hot)\ P(hot)$$

$$P(cloudy \wedge cold) = P(cloudy|cold)\ P(cold)$$

$$P(rainy \wedge hot) = P(rainy|hot)\ P(hot)$$

$$P(rainy \wedge cold) = P(rainy|cold)\ P(cold)$$

# JOINT PROBABILITY DISTRIBUTION

- You can also do inference
- $P(\text{hot} \mid \text{rainy}) = \dfrac{P(\text{hot} \wedge \text{rainy})}{P(\text{rainy})}$

| Weather | Temperature | Probability |
|---|---|---|
| sunny | hot | 150/365 |
| sunny | cold | 50/365 |
| cloudy | hot | 40/365 |
| cloudy | cold | 60/365 |
| rainy | hot | 5/365 |
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# Bayes' Theorem

# BAYES' THEOREM

- The chain rule and commutativity of conjunction ($h \wedge e$ is equivalent to $e \wedge h$) gives us:

$$P(h \wedge e) = P(h|e) \times P(e)$$
$$= P(e|h) \times P(h).$$

If $P(e) \neq 0$, divide the right hand sides by $P(e)$:

$$P(h|e) = \frac{P(e|h) \times P(h)}{P(e)}.$$

This is Bayes' theorem.

# EXAMPLE

In Orange County, 51% of the adults are males. One adult is randomly selected for a survey involving credit card usage.

a. Find the prior probability that the selected person is a male.
b. It is later learned that the selected survey subject was smoking a cigar. Also, 9.5% of males smoke cigars, whereas 1.7% of females smoke cigars (based on data from the Substance Abuse and Mental Health Services Administration). Use this additional information to find the probability that the selected subject is a male.

# EXAMPLE

- Let's use the following notation:

$M$ = male $\bar{M}$ = female (or not male)

$C$ = cigar smoker $\bar{C}$ = not a cigar smoker

a. Before using the information given in part b, we know only that 51% of the adults in Orange County are males, so the probability of randomly selecting an adult and getting a male is given by $P(M) = 0.51$.

# EXAMPLE

b. Based on the additional given information, we have the following:

$P(M) = 0.51$ because 51% of the adults are males

$P(\bar{M}) = 0.49$ because 49% of the adults are females (not males)

$P(C|M) = 0.095$ because 9.5% of the males smoke cigars (That is, the probability of getting someone who smokes cigars, given that the person is a male, is 0.095.)

$P(C|\bar{M}) = 0.017$. because 1.7% of the females smoke cigars (That is, the probability of getting someone who smokes cigars, given that the person is a female, is 0.017.)

# EXAMPLE

- Let's now apply Bayes' theorem by using the preceding formula with M in place of A, and C in place of B. We get the following result:

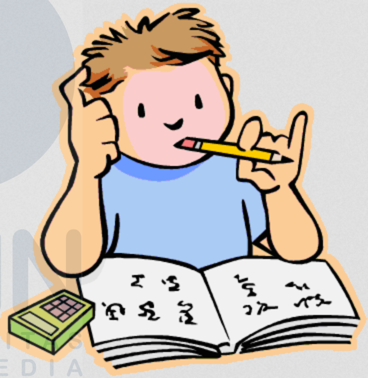
$$P(M|C) = \frac{P(M)P(C|M)}{[P(M)P(C|M)]+[P(\overline{M})P(C|\overline{M})]}$$

$$= \frac{0.51\ x\ 0.095}{[0.51\ x\ 0.095]+[0.49\ x\ 0.017]}$$

$$= 0.85329341$$

Before we knew that the survey subject smoked a cigar, there is a 0.51 probability that the survey subject is male (because 51% of the adults in Orange County are males). However, after learning that the subject smoked a cigar, we revised the probability to 0.853. There is a 0.853 probability that the cigar–smoking respondent is a male.

# PRACTICE



# PRACTICE 1

| | Positive Test Result (Pregnancy is indicated) | Negative Test Result (Pregnancy is not indicated) |
|---|---|---|
| **Subject is pregnant** | 80 | 5 |
| **Subject is not pregnant** | 3 | 11 |

a. If one of the 99 test subjects is randomly selected, what is the probability of getting a subject who is pregnant?

b. A test subject is randomly selected and is given a pregnancy test. What is the probability of getting a subject who is pregnant, given that the test result is positive?

# PRACTICE 1

| | Positive Test Result (Pregnancy is indicated) | Negative Test Result (Pregnancy is not indicated) |
|---|---|---|
| **Subject is pregnant** | 80 | 5 |
| **Subject is not pregnant** | 3 | 11 |

c. If one of the 99 test subjects is randomly selected, what is the probability of getting a subject who is not pregnant?

d. A test subject is randomly selected and is given a pregnancy test. What is the probability of getting a subject who is not pregnant, given that the test result is negative?

# PRACTICE 2

In a study of pleas and prison sentences, it is found that 45% of the subjects studied were sent to prison. Among those sent to prison, 40% chose to plead guilty. Among those not sent to prison, 55% chose to plead guilty.

a. If one of the study subjects is randomly selected, find the probability of getting someone who was not sent to prison.

b. If a study subject is randomly selected and it is then found that the subject entered a guilty plea, find the probability that this person was not sent to prison.

# PRACTICE 3

An aircraft emergency locator transmitter (ELT) is a device designed to transmit a signal in the case of a crash. The Altigauge Manufacturing Company makes 80% of the ELTs, the Bryant Company makes 15% of them, and the Chartair Company makes the other 5%. The ELTs made by Altigauge have a 4% rate of defects, the Bryant ELTs have a 6% rate of defects, and the Chartair ELTs have a 9% rate of defects (which helps to explain why Chartair has the lowest market share).

a. Find the probability of randomly selecting an ELT and getting one manufactured by the Chartair Company.

b. An ELT is randomly selected and tested. If the test indicates that the ELT is defective, find the probability that it was manufactured by the Chartair Company.

# PRACTICE 3

A = ELT manufactured by Altigauge

B = ELT manufactured by Bryant

C = ELT manufactured by Chartair

D = ELT is defective

$\bar{D}$ = ELT is not defective (or it is good)

# References

[1] S. J. Russell, P. Norvig, and E. Davis, **Artificial intelligence: a modern approach**, 3rd ed. Prentice Hall, 2010.

# Next Week

- Probabilistic Reasoning
- Bayesian Network
- Bayesian Network Inference

# Vision and Mission

## Vision

Becoming a leading Informatics Undergraduate Study Program that produces graduates with international insight who are competent in the field of computer science, have an entrepreneurial spirit, and have noble character.

## Mission

1. Organising learning with the best technology and curriculum and supported by professional teaching staff.
2. Carry out research activities in the field of informatics to advance information science and technology.
3. Carry out community service activities based on information technology and science in the context of practising informatics science and technology society.